

Homework 3

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Set up

```
# read in tidyverse for data wrangling and visualization
library(tidyverse)
# read in here for reproducible file paths
library(here)
# read in janitor for cleaning column names
library(janitor)

# read in kelp data from data folder
kelp <- read_csv(here("data", "temp-kelp.csv"))

# read in personal data from data folder
my_data <- read_csv(here("data", "MyData_Homework3.csv"))
```

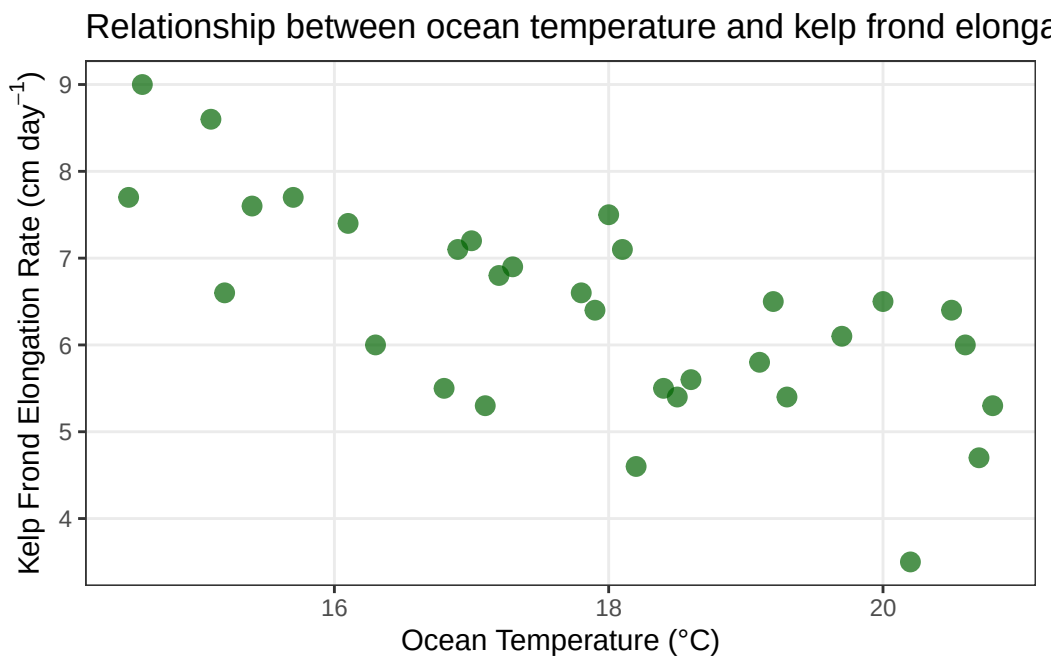
Problem 1: Giant kelp fronds

1a. An appropriate test

The two appropriate tests to determine the strength of the relationship between ocean temperature and giant kelp frond elongation rate are the Pearson correlation coefficient and the Spearman rank correlation coefficient. The Pearson correlation measures the strength and direction of a linear relationship between two continuous variables and assumes that both variables are normally distributed. The Spearman rank correlation is a non-parametric alternative that measures the strength of a monotonic relationship between ocean temperature and elongation rate without assuming normality, making it appropriate if the normality assumption for Pearson is violated.

1b. Create a visualization

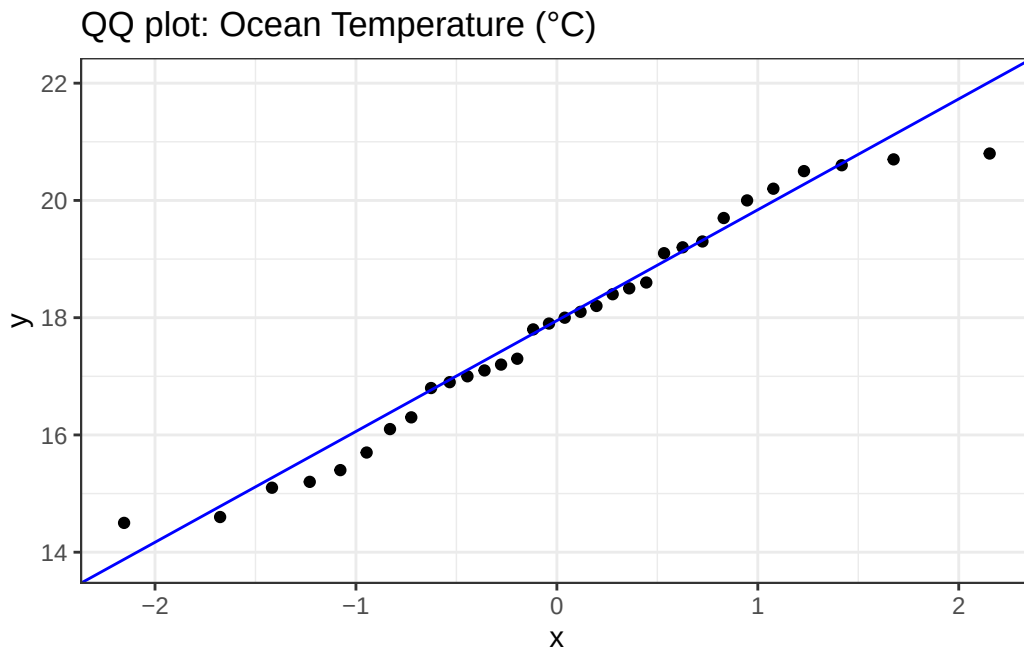
```
# create scatterplot of ocean temperature vs kelp frond elongation rate
ggplot(data = kelp, # use kelp data frame
       aes(x = temp_c, # ocean temperature on x-axis
           y = kelp_elong)) + # elongation rate on y-axis
geom_point(color = "darkgreen", # non-default point color
           size = 3,
           alpha = 0.7) + # slight transparency on points
labs(x = "Ocean Temperature (°C)", # label x-axis with units
     y = expression("Kelp Frond Elongation Rate (cm day-1)"),
     title = "Relationship between ocean temperature and kelp frond elongation rate") +
theme_bw() + # non-default theme
theme(panel.grid.minor = element_blank()) # remove minor gridlines to reduce clutter
```



1c. Check your assumptions and run your test

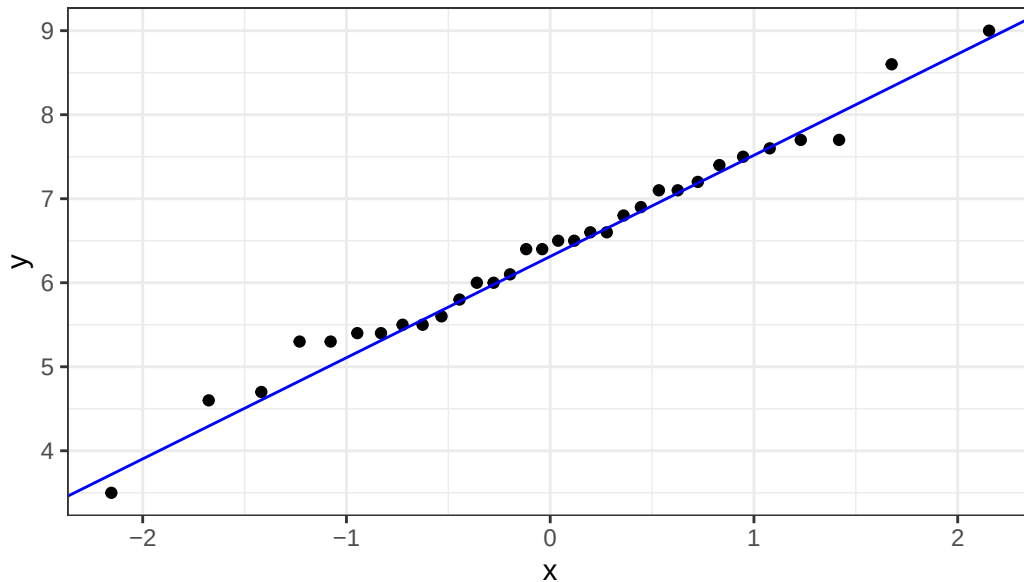
```
# check normality of ocean temperature using a QQ plot
ggplot(data = kelp,
       aes(sample = temp_c)) + # check temperature variable
```

```
geom_qq() + # plot sample quantiles against theoretical quantiles
geom_qq_line(color = "blue") + # add reference line for normality
labs(title = "QQ plot: Ocean Temperature (°C)") +
theme_bw() # non-default theme
```



```
# check normality of kelp elongation rate using a QQ plot
ggplot(data = kelp,
       aes(sample = kelp_elong)) + # check elongation rate variable
geom_qq() + # plot sample quantiles against theoretical quantiles
geom_qq_line(color = "blue") + # add reference line for normality
labs(title = expression("QQ plot: Kelp Frond Elongation Rate (cm day"^-1*")")) +
theme_bw() # non-default theme
```

QQ plot: Kelp Frond Elongation Rate (cm day^{-1})



```
# run Pearson correlation test between ocean temperature and kelp elongation rate
cor.test(kelp$temp_c, # predictor variable: ocean temperature
         kelp$kelp_elong, # response variable: kelp frond elongation rate
         method = "pearson") # specify Pearson method for linear relationship
```

Pearson's product-moment correlation

```
data: kelp$temp_c and kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
      cor
-0.6877489
```

Both variables were assessed for normality using QQ plots. Ocean temperature ($^{\circ}\text{C}$) appeared approximately normally distributed, as the points followed the reference line closely with only minor deviations at the tails. Kelp frond elongation rate also appeared approximately normally distributed, with points tracking the reference line well. Because both variables met the normality assumption and the relationship appeared linear in the scatterplot, I proceeded with the Pearson correlation coefficient.

1d. Results communication

To evaluate the strength of the relationship between ocean temperature and giant kelp frond elongation rate, I used a Pearson correlation coefficient because both variables are continuous and approximately normally distributed (confirmed via QQ plots), and the scatterplot suggested a linear relationship. The results indicate a statistically significant, moderate negative relationship between ocean temperature and kelp frond elongation rate (Pearson's $r = -0.68$, $t(30) = -5.19$, $p < 0.001$), meaning that as ocean temperature increases, kelp frond elongation rate tends to decrease.

1e. Test implications

Our results indicate that higher ocean temperatures are associated with significantly lower giant kelp frond elongation rates, suggesting that ongoing ocean warming due to climate change may slow kelp growth. This is critical for our team to understand because reduced growth rates could diminish kelp forest biomass and structural complexity over time, threatening the many species that depend on these ecosystems. As ocean temperatures continue to rise, we should prioritize monitoring kelp growth rates in warmer regions as early indicators of broader kelp forest decline.

1f. Double check your own work

```
# run Spearman rank correlation as the alternative non-parametric test
# does not assume normality, measures monotonic relationship
cor.test(kelp$temp_c, # predictor variable: ocean temperature
         kelp$kelp_elong, # response variable: kelp frond elongation rate
         method = "spearman") # specify Spearman method
```

Spearman's rank correlation rho

```
data: kelp$temp_c and kelp$kelp_elong
S = 9216.1, p-value = 1.29e-05
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
-0.6891684
```

Both the Pearson correlation ($r = -0.69$, $t(30) = -5.19$, $p < 0.001$) and the Spearman rank correlation ($\rho = -0.69$, $S = 9216.1$, $p < 0.001$) led to the same decision to reject the null hypothesis that there is no relationship between ocean temperature and giant kelp frond elongation rate. Both tests revealed a strong negative relationship of nearly identical magnitude, meaning that both would lead to the same interpretation that higher ocean temperatures are associated with lower kelp frond elongation rates. The close agreement between the two tests strengthens confidence in the conclusion, as the Pearson result indicates the relationship is linear while the Spearman result confirms it is also monotonic.

Problem 2. Personal Data

2a. Updating visualizations

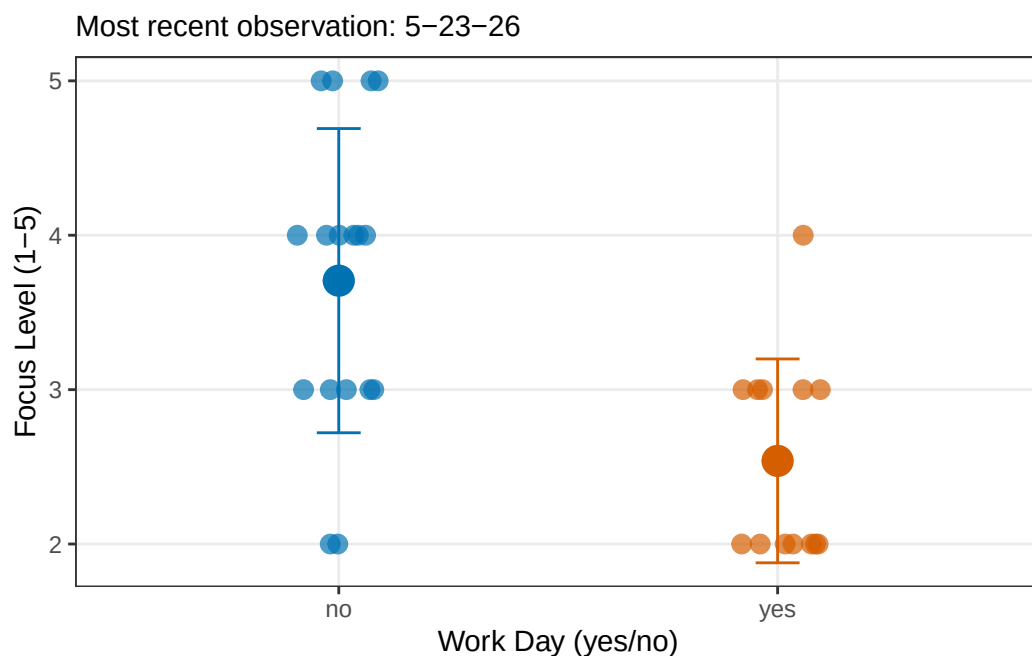
```
# summarize mean focus level and SD by work day
my_data_summary <- my_data |>
  group_by(work_day) |> # group by whether it was a work day
  summarize(
    mean_focus = mean(focus_level), # calculate mean focus level per group
    sd_focus = sd(focus_level) # calculate standard deviation per group
  )
```

```
# plot focus level by work day with raw data, mean, and error bars
ggplot() +
  geom_jitter(data = my_data, # plot raw individual data points
    aes(x = work_day,
        y = focus_level,
        color = work_day),
    width = 0.1, # jitter horizontally only
    height = 0,
    size = 3,
    alpha = 0.7) + # slight transparency
  geom_point(data = my_data_summary, # plot group mean as larger point
    aes(x = work_day,
        y = mean_focus,
        color = work_day),
    size = 5) +
  geom_errorbar(data = my_data_summary, # plot SD as error bars
    aes(x = work_day,
        ymin = mean_focus - sd_focus, # lower bound of SD
        ymax = mean_focus + sd_focus, # upper bound of SD
```

```

    color = work_day),
    width = 0.1) +
scale_color_manual(values = c("yes" = "#D55E00", # non-default colors
                              "no" = "#0072B2")) +
labs(x = "Work Day (yes/no)", # label x-axis
     y = "Focus Level (1-5)", # label y-axis with units
     subtitle = paste("Most recent observation:", tail(my_data$date, 1))) +
# most recent date))) + # most recent date
theme_bw() + # non-default theme
theme(legend.position = "none", # remove redundant legend
      panel.grid.minor = element_blank()) # reduce visual clutter

```

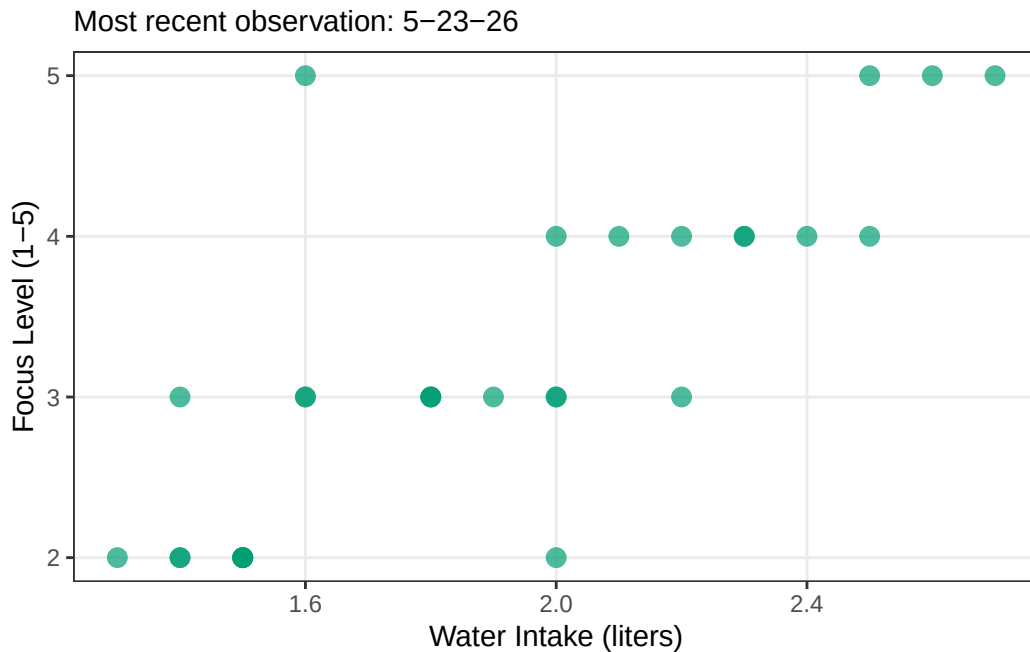


```

# plot water intake vs focus level as scatterplot
ggplot(data = my_data,
       aes(x = water_intake_L, # water intake on x-axis
           y = focus_level)) + # focus level on y-axis
geom_point(color = "#009E73", # non-default color
           size = 3,
           alpha = 0.7) + # slight transparency
labs(x = "Water Intake (liters)", # label x-axis with units
     y = "Focus Level (1-5)", # label y-axis with units
     subtitle = paste("Most recent observation:", tail(my_data$date, 1))) +

```

```
theme_bw() + # non-default theme
theme(panel.grid.minor = element_blank()) # reduce visual clutter
```



2b. Captions

Figure 1. Focus level (1–5 scale) on work days versus non-work days, based on 13 days of personal observations. Points show individual daily observations, larger points show group means, and error bars represent ± 1 standard deviation. Focus level was slightly lower on work days compared to non-work days, with greater variability on work days. Figure 2. Relationship between daily water intake (liters) and focus level (1–5 scale) across 13 days of personal observation. Each point represents one day. A positive association between water intake and focus level is visible, suggesting that higher water intake may be associated with higher focus, though the sample size is small and this pattern should be interpreted cautiously.

Problem 3. Affective visualization

3a. Description

An affective visualization for my personal data could take the form of a hand-drawn weekly calendar where each day is represented as a water droplet shape, with the size of the droplet

corresponding to how much water I drank that day and the color (ranging from pale yellow to deep blue) representing my focus level for that day. Days where I worked could be marked with a small sun icon, while rest days could have a moon. This format would make the data feel personal and embodied — the viewer would literally see how “full” or “energized” each day felt, connecting hydration and mental energy in an intuitive visual language that goes beyond what a standard chart could convey.

3b. Sketch

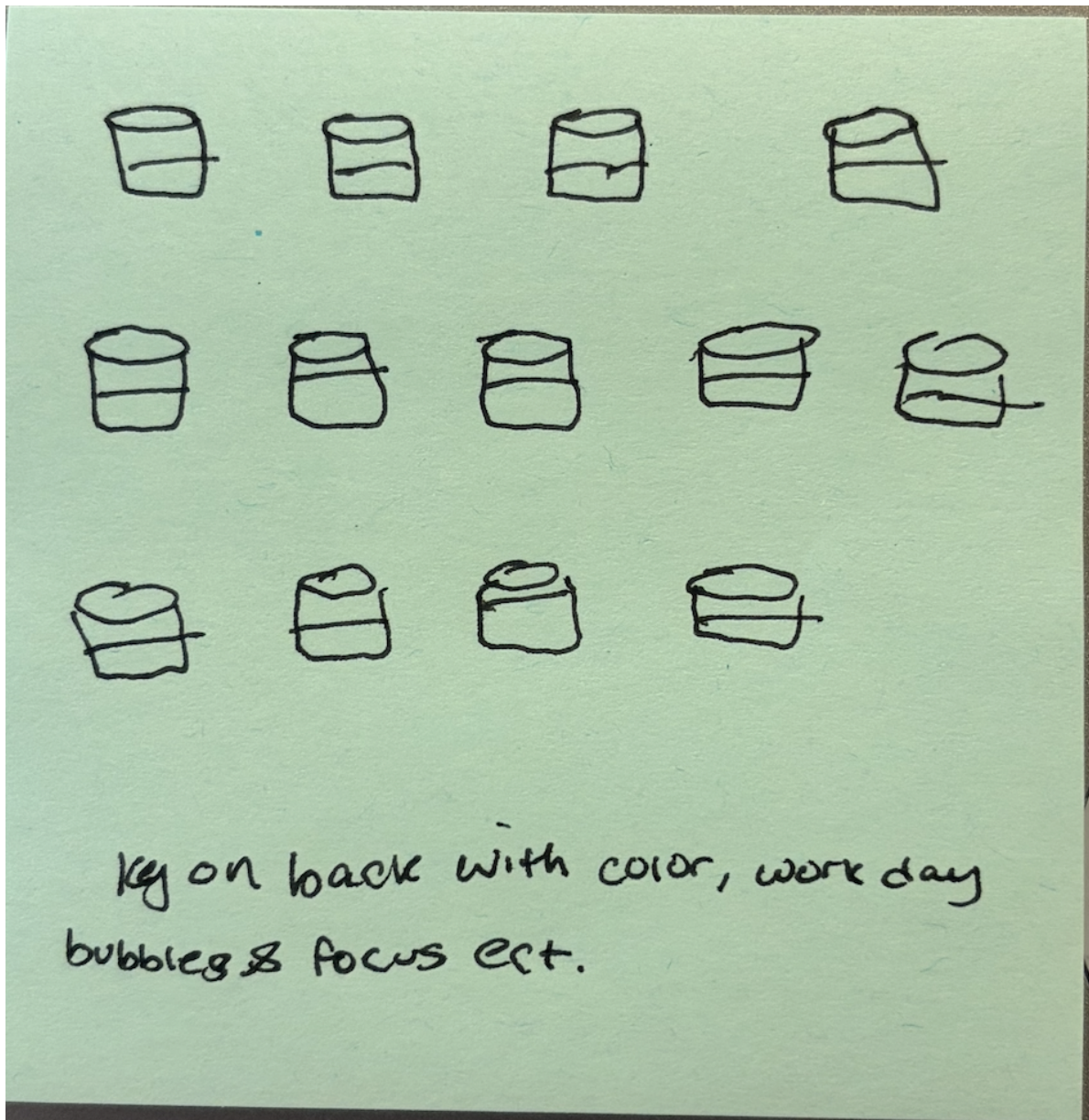


Figure 1: Initial Sketch for drawing

3c. Draft

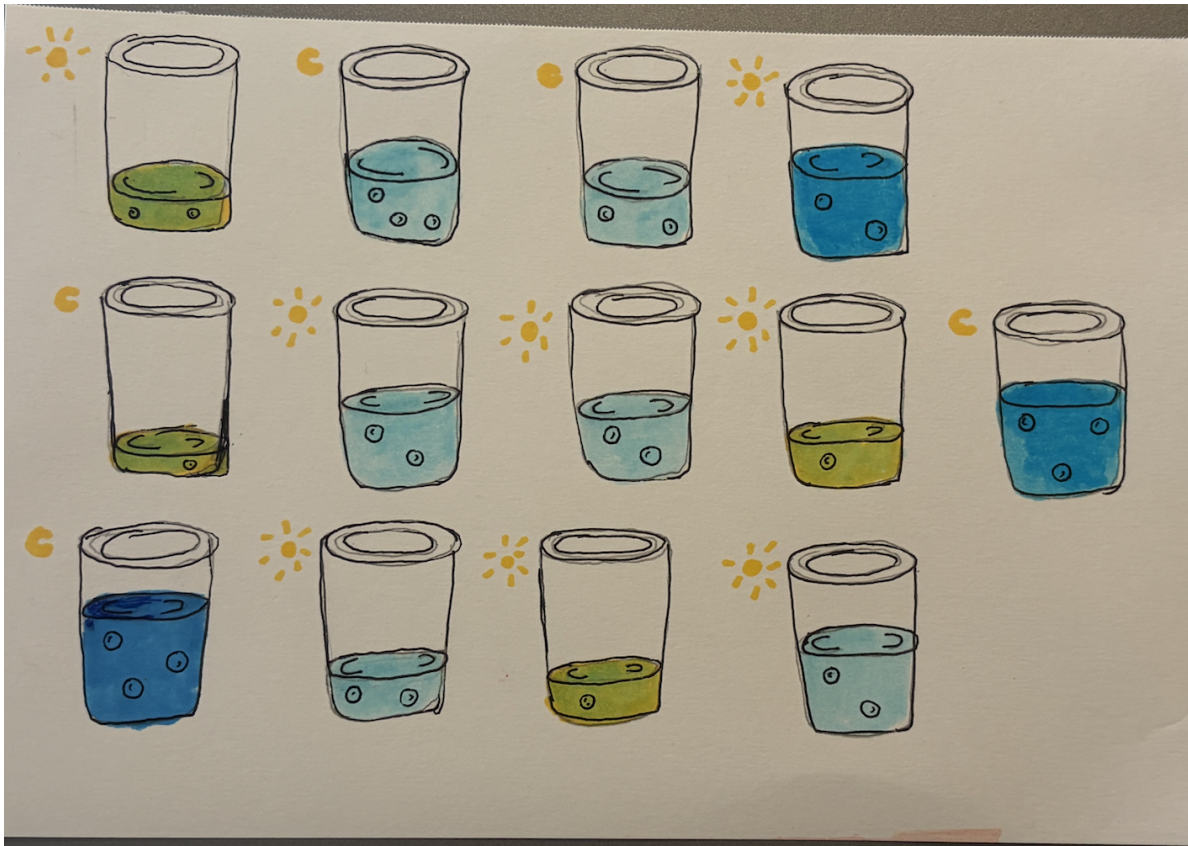


Figure 2: Draft of effective visualization, this is for only 13 days but the full one would be 30

3d. Artist Statement

Content: This piece shows 13 days of personal observations tracking daily water intake and focus level, with additional variables including number of study sessions and whether the day was a work day or rest day, exploring whether staying hydrated is associated with feeling more mentally focused. **Influences:** I was directly influenced by Stefanie Posavec and Giorgia Lupi's Dear Data project, which encodes personal behavioral data into dense hand-drawn postcards using repeated simple shapes with a legend, making data feel intimate and personal rather than clinical. **Form:** The piece is a hand-drawn postcard created with colored pencils and pen on paper, designed to be small and personal in the style of Dear Data. **Process:** I first mapped each variable to a visual element — water intake to fill level, focus to color, study sessions to bubbles, and work day to sun or moon symbols. I then sketched all 13 glasses in pencil, added the water fill and color for each day using the data table, drew the bubbles and symbols, and finished with a hand-drawn legend in the corner.

3e. Slides

[View slides here](#)

Problem 4. Statistical critique

4a. Revisit and summarize

The authors of Estrada-Araoz et al. 2023 use the Pearson correlation coefficient to address their main research question of whether environmental education is significantly related to pro-environmental behavior in regular basic education students in the Peruvian Amazon. The response variable is pro-environmental behavior score and the predictor variable is environmental education score, both measured via questionnaire. The Pearson correlation tests whether there is a significant linear relationship between the two scores, and a significant result ($p < 0.05$) would indicate that students who perceive their environmental education as more adequate also tend to engage in pro-environmental behaviors more frequently.

Table 4. Correlation matrix between variables and dimensions.

Variables and Dimensions	1	2	3	4	5	6	7	8
1. Environmental education	1							
2. Cognitive aspect	0.893 *	1						
3. Affective aspect	0.892 *	0.835 *	1					
4. Behavioral aspect	0.879 *	0.812 *	0.803 *	1				
5. Pro-environmental behavior	0.877 *	0.799 *	0.798 *	0.737 *	1			
6. Recycling and reuse	0.863 *	0.811 *	0.760 *	0.762 *	0.915 *	1		
7. Resource saving	0.889 *	0.764 *	0.773 *	0.740 *	0.920 *	0.879 *	1	
8. Environmental purchasing	0.894 *	0.752 *	0.763 *	0.768 *	0.900 *	0.855 *	0.894 *	1

* The correlation is significant at the 0.01 level (bilateral).

Figure 3: Table 4 from Estrada-Araoz et al. 2023 showing Pearson correlation coefficients between environmental education and pro-environmental behavior variables.

4b. Visual clarity

Since the authors presented their statistics in a correlation matrix table rather than a figure, I will evaluate how clearly the table represents the data underlying the tests. Table 4 clearly organizes all variables and their subdimensions as both rows and columns, making every pairwise Pearson correlation coefficient easy to locate by finding the intersection of any two variables. The authors include asterisks to denote statistical significance at $p < 0.05$, which helps the reader quickly identify which relationships are meaningful without needing to interpret raw p-values. However the table does not include confidence intervals, sample sizes for individual correlations, or p-values themselves, which would have allowed readers to more fully evaluate the precision and strength of each estimate.

4c. Aesthetic clarity

The correlation matrix in Table 4 handles visual clutter reasonably well, there are no shaded cells, excessive borders, or redundant decorative elements, giving it a relatively high data-to-ink ratio. The asterisks used to denote statistical significance add meaningful information without adding unnecessary visual noise. However, the table does not use any bolding or italics to draw the reader's eye to the most important correlations, such as the $r = 0.877$ between overall environmental education and pro-environmental behavior, meaning the reader must scan every cell to find the key findings rather than being guided to them visually. Additionally, the lower triangle of the matrix is a mirror of the upper triangle and adds redundant ink without adding new information.

4d. Recommendations

I would recommend replacing the correlation matrix table with a heatmap figure, where the x- and y-axes list all eight variables and each cell is filled with a color gradient (e.g., white to dark teal) proportional to the Pearson r value. Significant correlations could be marked with an asterisk inside the cell. This would make it immediately visually clear which variable pairs have the strongest relationships without requiring the reader to scan rows of numbers, and the color encoding would allow at-a-glance comparison across all pairs. Including a color scale legend would make the figure self-explanatory, improving both clarity and accessibility.